Paper Simulation II: shrinkage

N = 40 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.57	29.52	6.72/7.25/4.37	6.72/7.25/4.37	7.55/8.34/4.99
Scenario II: T1 vs. T0	24.78	52.65	26.20/27.59/20.81	41.49/42.76/36.13	27.88/29.04/22.58
Scenario II: T2 vs. T0	44.72	73.63	45.49/47.09/39.14	58.41/59.88/51.83	48.22/49.99/41.69
Scenario II: T3 vs. T0	66.82	88.34	65.98/67.84/60.21	73.88/75.29/68.41	69.32/70.98/63.48
Scenario II: T4 vs. T0	84.64	96.03	83.49/84.93/79.02	87.23/88.13/83.06	86.09/87.51/82.18
Scenario II: T5 vs. T0	93.53	98.88	92.51/93.52/90.14	93.79/94.63/91.49	94.04/94.90/92.20
Scenario II: joint power	16.80	44.64	17.79/18.93/13.38	36.51/37.84/31.34	19.61/20.68/15.04

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Paper Simulation II: shrinkage

N = 80 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.43	29.21	5.53/6.11/4.32	5.53/6.11/4.32	6.59/7.26/5.18
Scenario II: T1 vs. T0	52.75	79.39	50.64/52.72/47.80	71.38/72.88/69.08	53.52/55.16/50.78
Scenario II: T2 vs. T0	80.69	94.68	79.44/80.40/77.42	88.54/89.10/87.33	81.64/82.78/79.62
Scenario II: T3 vs. T0	94.96	99.20	94.38/94.82/93.43	96.75/97.09/96.28	95.37/95.86/94.57
Scenario II: T4 vs. T0	99.27	99.93	99.00/99.13/98.79	99.38/99.43/99.27	99.31/99.37/99.14
Scenario II: T5 vs. T0	99.90	99.99	99.87/99.89/99.87	99.90/99.93/99.89	99.93/99.93/99.90
Scenario II: joint power	48.36	77.16	46.55/48.37/43.64	69.49/70.94/67.41	49.40/51.22/46.69

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Paper Simulation II: non−shrinkage

N = 40 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.81	29.76	6.72/7.25/4.37	6.72/7.25/4.37	7.76/8.37/5.19
Scenario II: T1 vs. T0	25.64	52.89	26.20/27.59/20.81	41.49/42.76/36.13	28.33/29.37/23.06
Scenario II: T2 vs. T0	44.48	73.45	45.49/47.09/39.14	58.41/59.88/51.83	48.28/49.74/41.85
Scenario II: T3 vs. T0	65.89	87.94	65.98/67.84/60.21	73.88/75.29/68.41	68.83/70.44/63.41
Scenario II: T4 vs. T0	83.31	95.82	83.49/84.93/79.02	87.23/88.13/83.06	85.58/86.85/81.43
Scenario II: T5 vs. T0	92.67	98.71	92.51/93.52/90.14	93.79/94.63/91.49	93.56/94.54/91.71
Scenario II: joint power	16.90	44.49	17.79/18.93/13.38	36.51/37.84/31.34	19.70/20.81/15.33

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Paper Simulation II: non−shrinkage

N = 80 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.47	28.95	5.53/6.11/4.32	5.53/6.11/4.32	6.52/7.22/5.16
Scenario II: T1 vs. T0	53.12	79.45	50.64/52.72/47.80	71.38/72.88/69.08	53.61/55.39/50.96
Scenario II: T2 vs. T0	80.64	94.52	79.44/80.40/77.42	88.54/89.10/87.33	81.55/82.53/79.47
Scenario II: T3 vs. T0	94.43	99.19	94.38/94.82/93.43	96.75/97.09/96.28	95.19/95.68/94.35
Scenario II: T4 vs. T0	99.09	99.93	99.00/99.13/98.79	99.38/99.43/99.27	99.25/99.29/99.07
Scenario II: T5 vs. T0	99.88	99.99	99.87/99.89/99.87	99.90/99.93/99.89	99.92/99.91/99.90
Scenario II: joint power	48.34	77.14	46.55/48.37/43.64	69.49/70.94/67.41	49.51/51.28/46.67

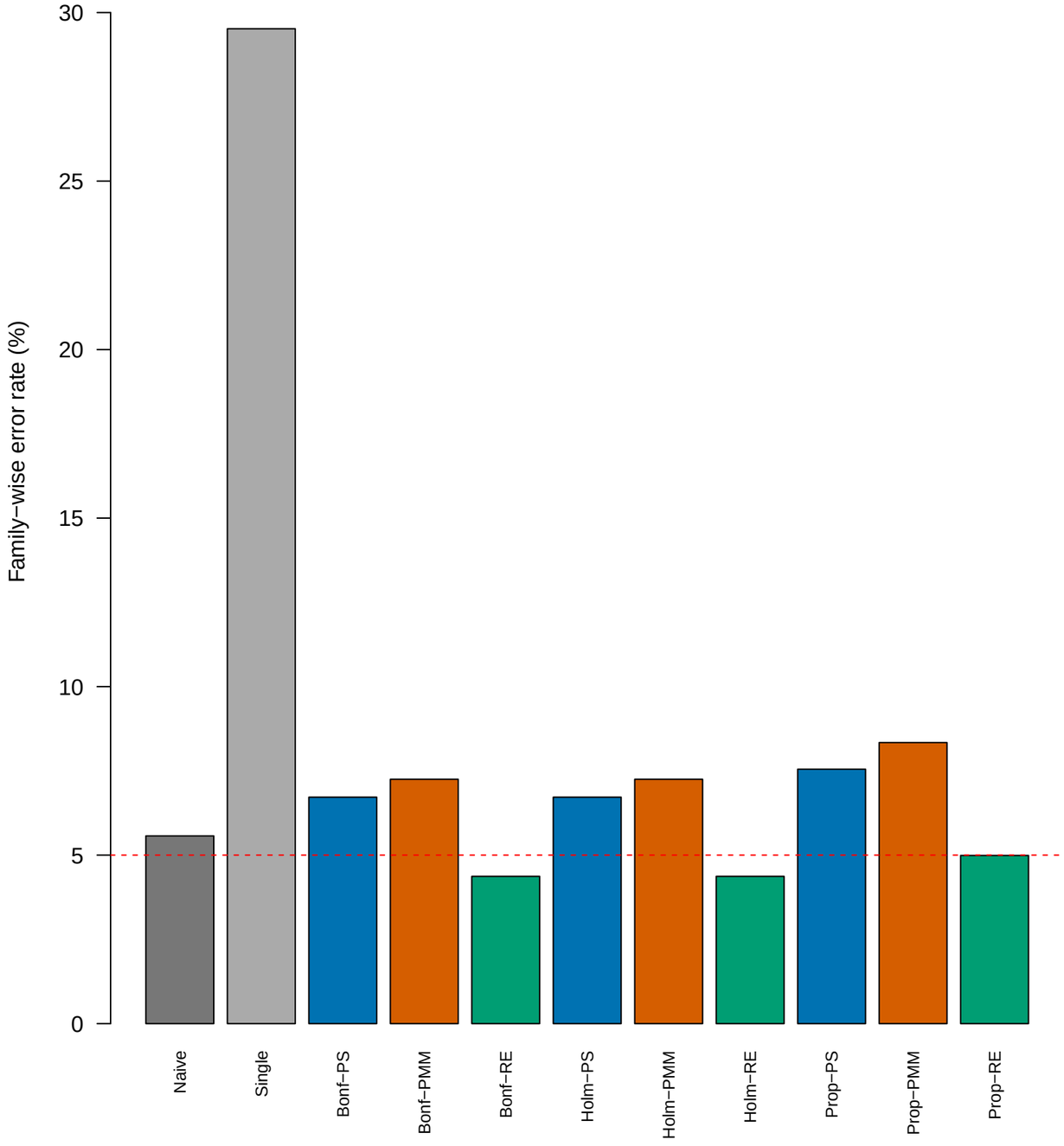
All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all five nulls in Scenario II.

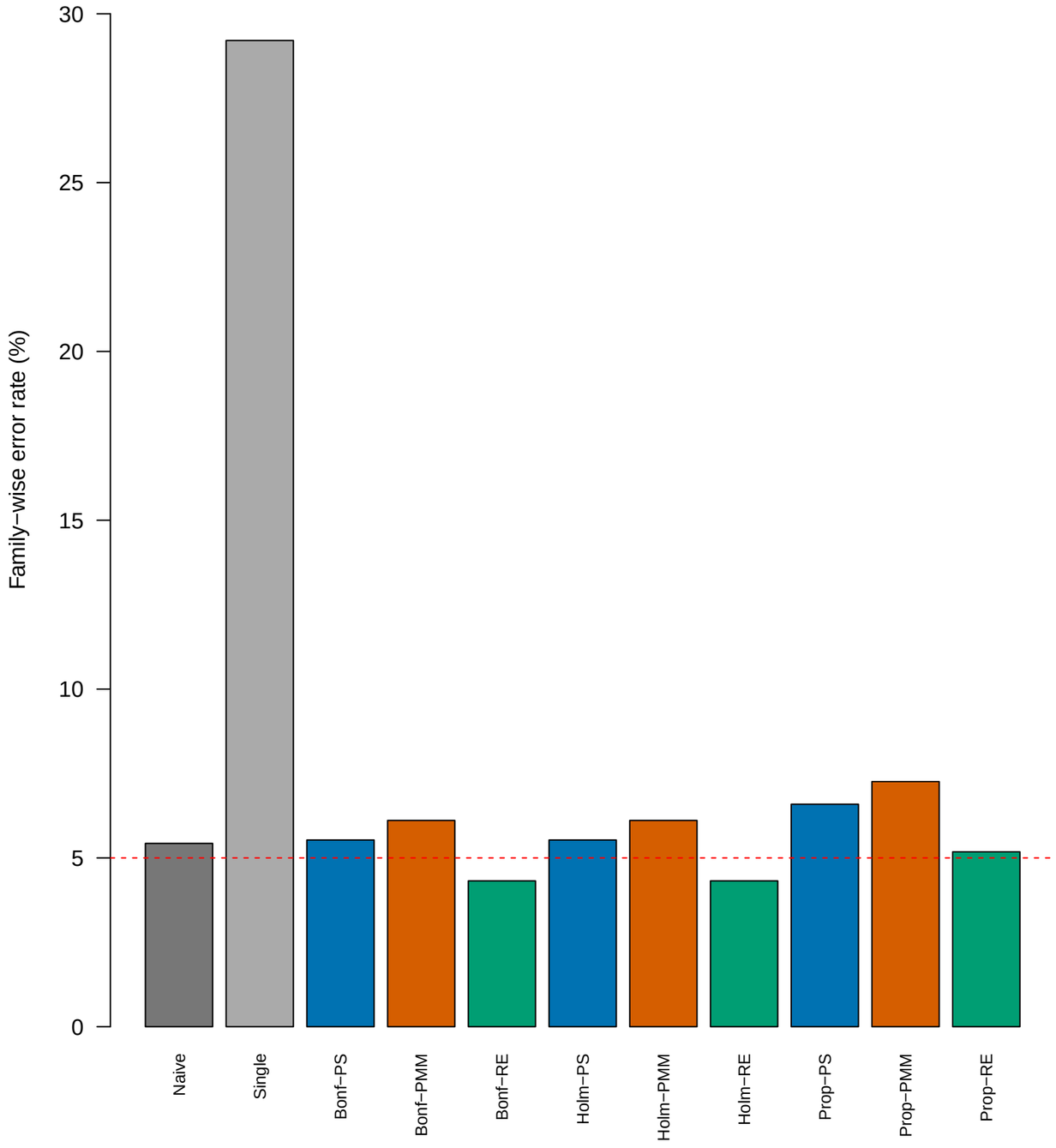
Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Simulation II: shrink − Scenario I

N = 40



N = 80



Simulation II: shrink − Scenario II

N = 40

N = 80

Joint power (%)

100

80

60

40

20

0

Naive

Single

Bonf−PS

Bonf−PMM

Bonf−RE

Holm−PS

Holm−PMM

Holm−RE

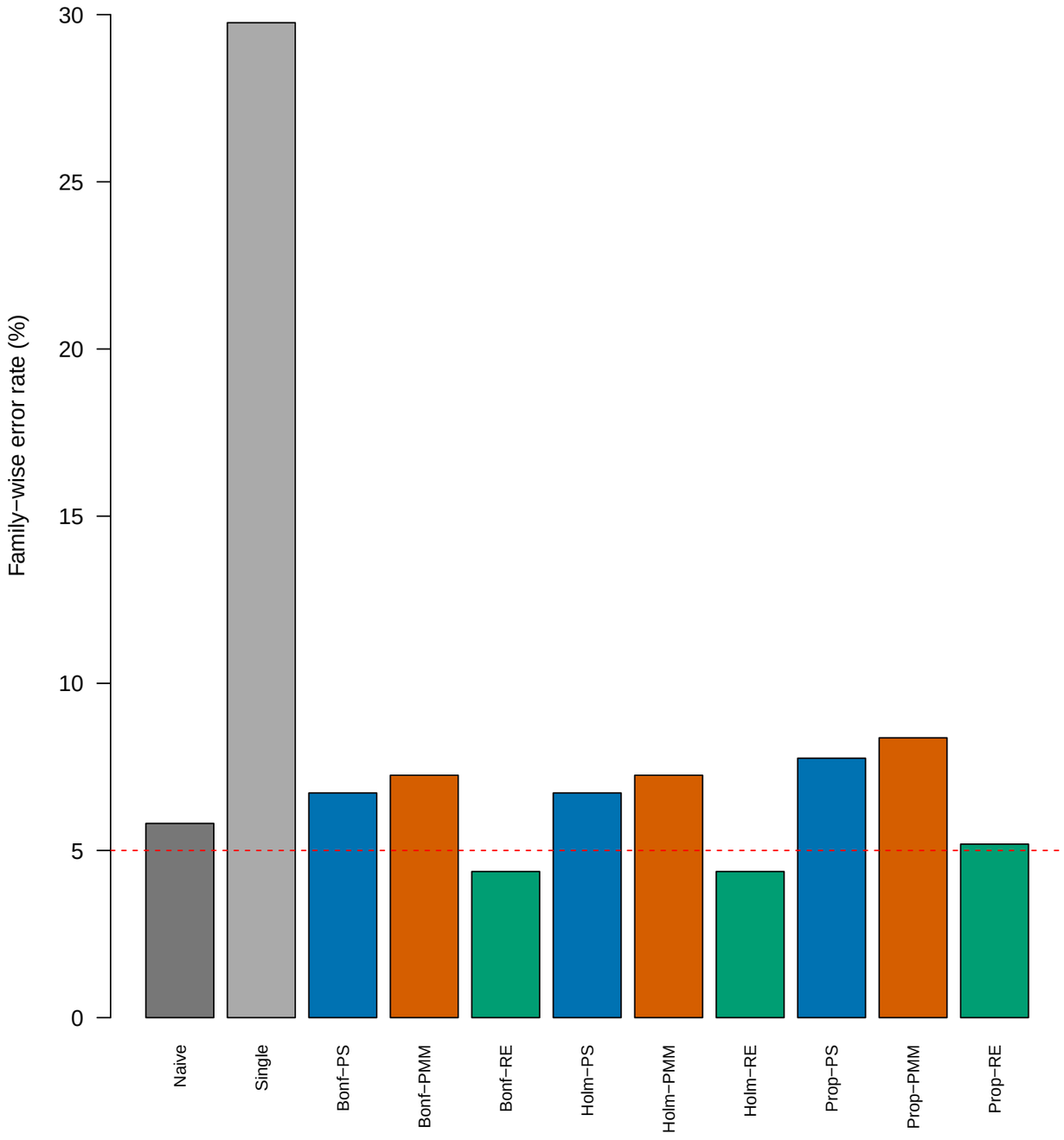
Prop−PS

Prop−PMM

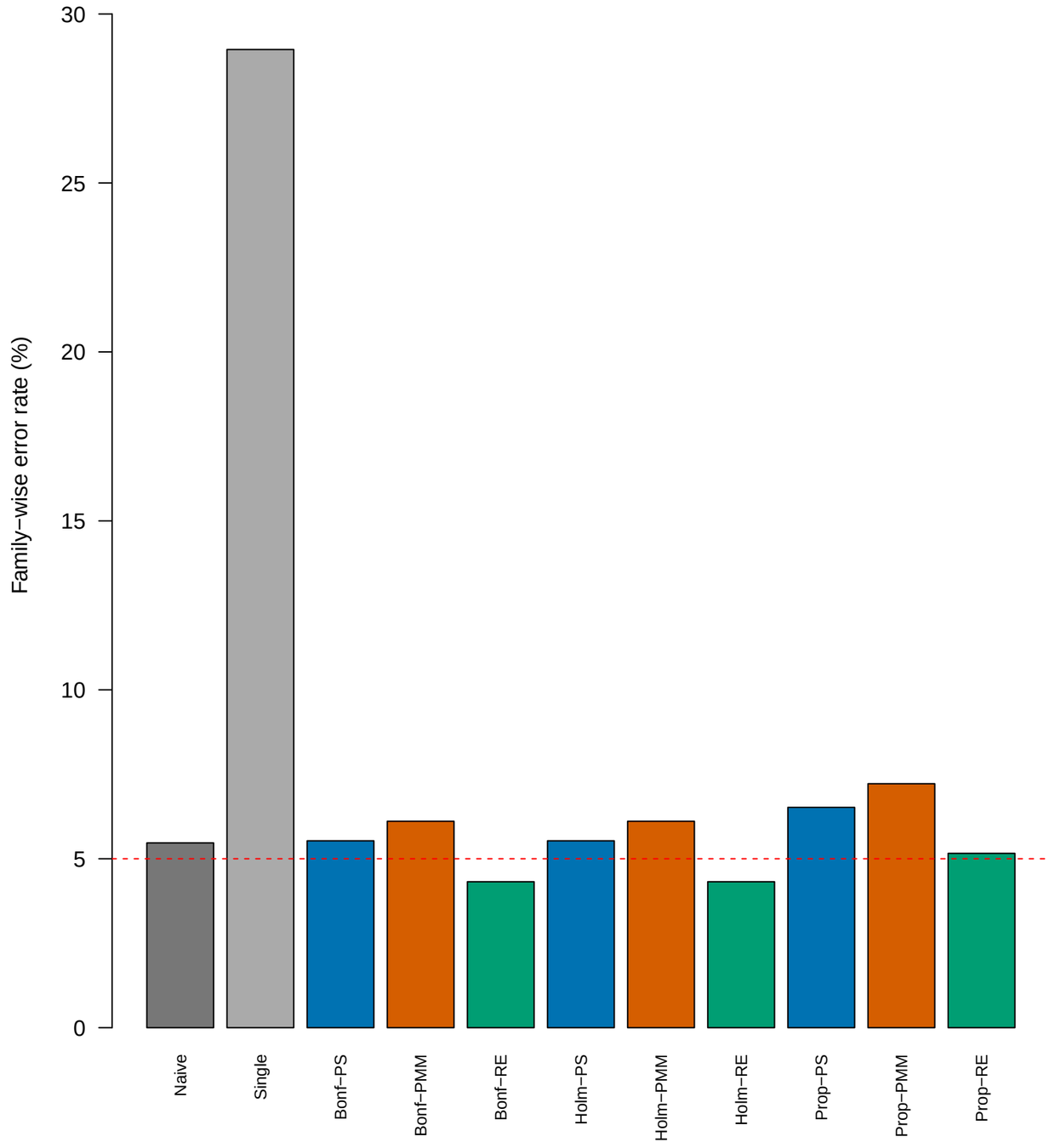
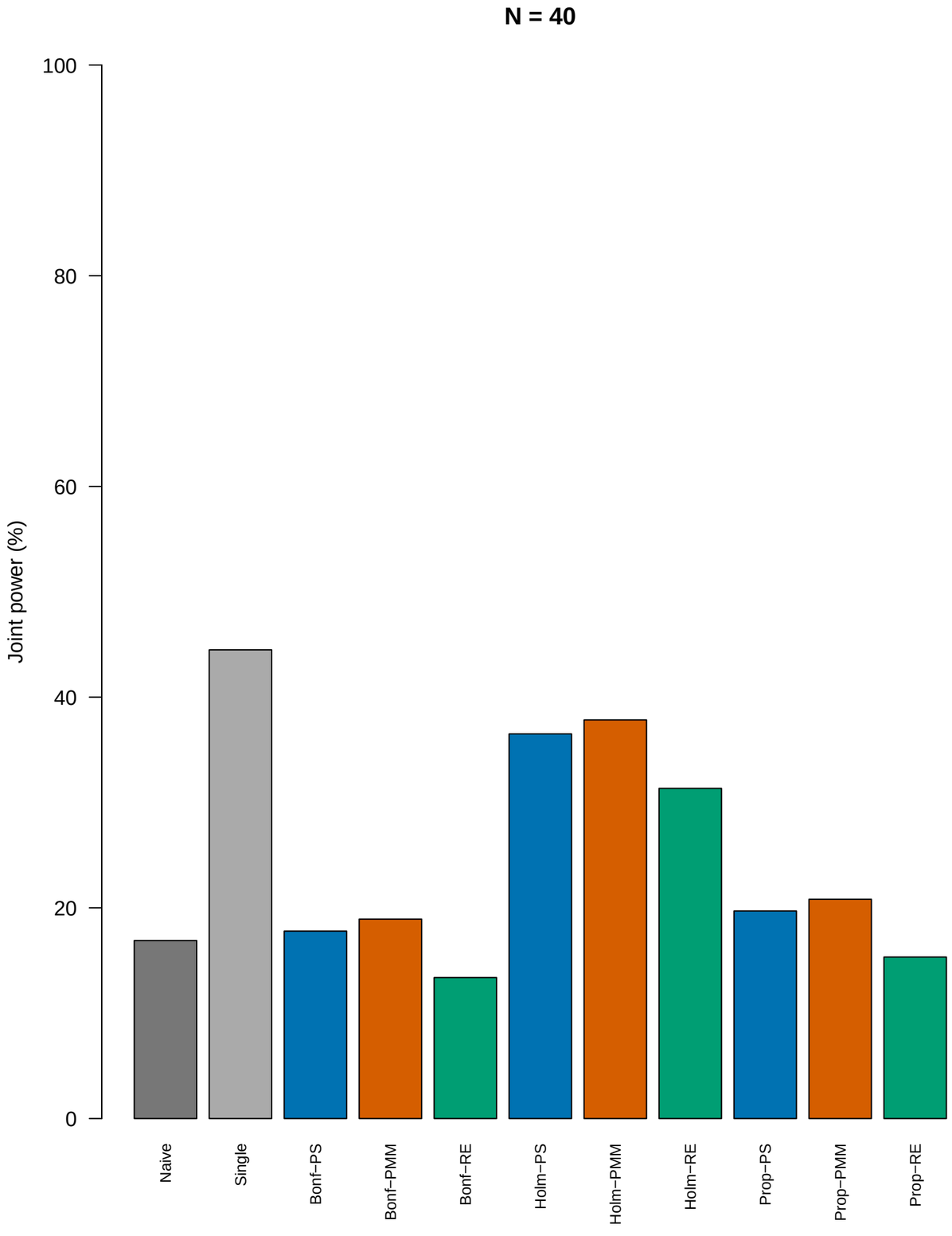
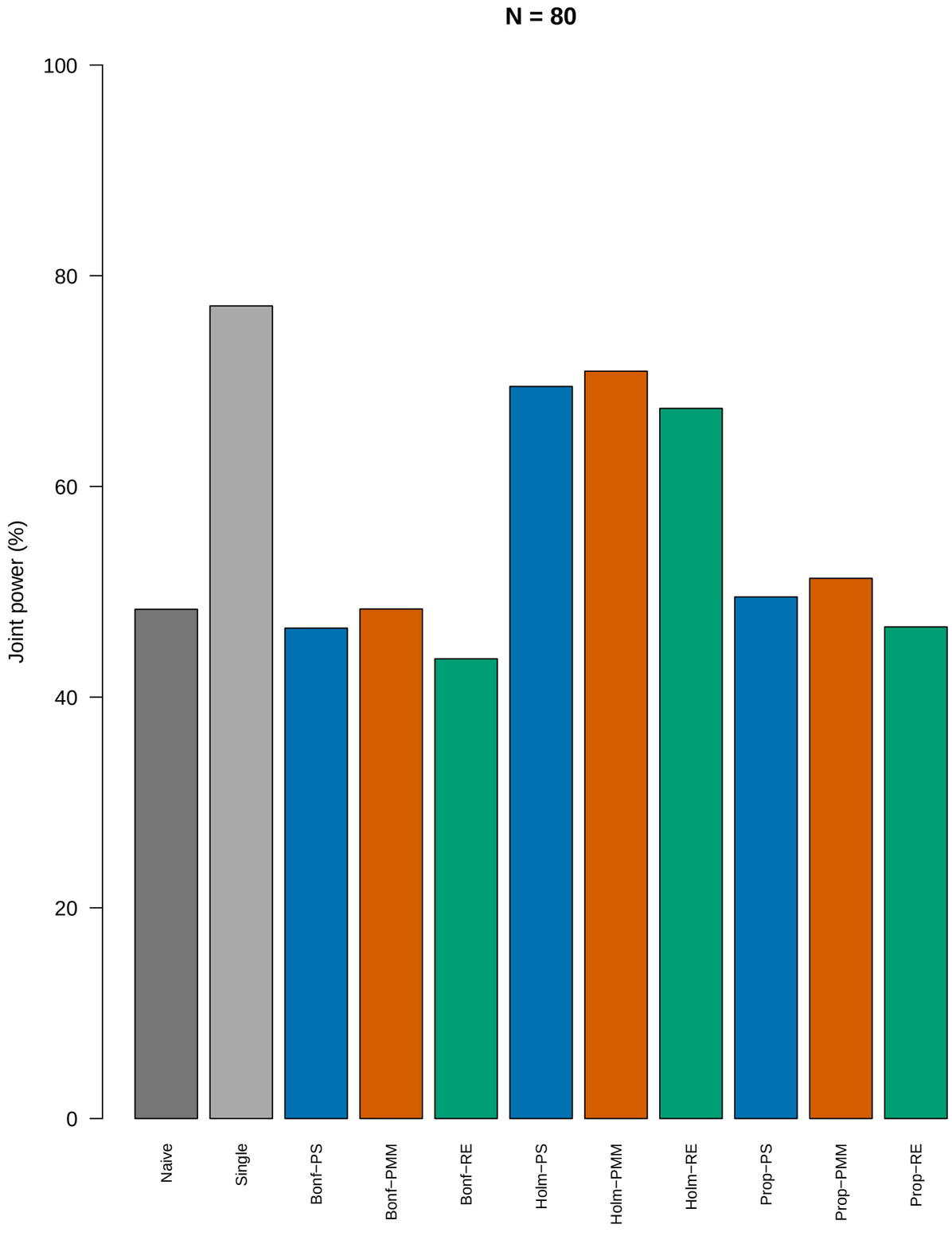
Prop−RE

Simulation II: nonshrink − Scenario I

N = 40



N = 80



Simulation II: nonshrink − Scenario II